Unit: Unit 8: Quadratics
Topic: Converting Forms (Completing the Square)

Name: ________________
Date: ____________

Quiz (Habanero)

Q1. The quadratic $x^2 + 8x + 16$ can be rewritten in vertex form as:
(A) $(x + 4)^2$
(B) $(x - 4)^2$
(C) $(x + 2)^2 + 4$
(D) $(x - 2)^2 - 4$
(E) $x^2 + 4x + 4$

Q2. To convert the quadratic $x^2 - 10x$ to vertex form, we need to add and subtract:
(A) 25
(B) 10
(C) 5
(D) 15
(E) 20

Q3. If a quadratic equation is in the form ax^2+bx+c, the vertex form is $a(x - h)^2 + k$. What is h in terms of a, b, and c?
(A) $-\frac{b}{2a}$
(B) $\frac{b}{2a}$
(C) $-\frac{c}{2a}$
(D) $\frac{c}{2a}$
(E) $-\frac{2a}{b}$

Q4. What is the vertex of the parabola given by the equation $y = x^2 - 6x + 5$?
(A) $(3, -4)$
(B) $(3, 4)$
(C) $(-3, 4)$
(D) $(-3, -4)$
(E) $(0, 5)$

Q5. Suppose the temperature T (in °C) of a certain location is modeled by the quadratic $T = x^2 - 4x - 5$. What is the vertex of this parabola?
(A) $(2, -9)$
(B) $(2, 9)$
(C) $(-2, 9)$
(D) $(-2, -9)$
(E) $(0, -5)$

Q6. If a quadratic function $f(x) = ax^2 + bx + c$ has its vertex at (h, k), what is $f(h)$?
(A) a
(B) b
(C) c
(D) k
(E) h

Q7. What is the equation of the parabola with vertex $(1, 2)$ and passing through the point $(0, 3)$?
(A) $y = (x - 1)^2 + 2$
(B) $y = -(x - 1)^2 + 2$
(C) $y = (x + 1)^2 - 2$
(D) $y = x^2 + 2x + 1$
(E) $y = x^2 - 2x + 3$

Q8. Given the equation $y = 2x^2 - 4x - 3$, what is the y-coordinate of the vertex?
(A) -7
(B) -5
(C) -3
(D) 1
(E) 5

Open Ended Questions (FRQ)

Q9. A quadratic function models the depth of water D (in meters) at a certain location in a river, given by $D = x^2 - 10x + 25$, where x is the distance (in km) from the river bank. Find the vertex of the parabola and interpret its meaning in this context. Show all work and provide a clear explanation.

Q10. Complete the square to rewrite the quadratic $x^2 + 12x + 20$ in vertex form. Then, graph the parabola and identify the vertex and x-intercepts. Show all work and provide a clear explanation.

Chef's Tips

- Emphasize the importance of completing the square to convert standard form to vertex form.
- Remind students to identify the vertex of a parabola from its equation in vertex form.
- Encourage students to interpret the vertex in context for word problems.